

Lab 7: Recycle Bin and Anti-forensics

What You Need for this lab

- Install Virtualbox : <https://www.virtualbox.org/wiki/Downloads>
- Install Kali 2021.4. : <https://old.kali.org/kali-images/kali-2021.4/>
 - Notes: Suggest You configure the disk size of Kali VM 80G because the size of each leakage cases image is 30G+
- Image “cfreds_2015_data_leakage_pc.dd” from Lab 5

1. Recycle Bin

Step 1.

- Examine ‘Recycle Bin’ data in PC.

```
root@kali:~/lab# /bin/bash 115x21
root@kali:~/lab# fls -rd -o 206848 cfreds_2015_data_leakage_pc.dd | grep -Ei "Recycle" | head
r/- * 0:      $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/desktop.ini
-/r * 74311-128-1:  $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$I55Z163
-/r * 74312-128-1:  $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$IXWGVWC
-/r * 74313-128-1:  $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$I40295N
-/r * 74395-128-1:  $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$I9M7UMY
-/r * 74398-128-1:  $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$RIQGWTI.ini
-/r * 74418-128-1:  $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$RJEMT64.exe
-/r * 74418-128-4:  $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$RJEMT64.exe
-/r * 74620-128-1:  $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$RKXD1U3.jpg
-/r * 74625-128-1:  $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$RI3FM2A.jpg
root@kali:~/lab#
```

informant SID = 1000 (see question 6)

Step 2.

- Use *testdisk* to Recover *\$Recycle.bin*

Recover the image

```
root@kali:~/lab# /bin/bash 1
root@kali:~/lab# testdisk cfreds_2015_data_leakage_pc.dd
```

```
/bin/bash 65x19
TestDisk 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media (use Arrow keys, then press Enter):
>Disk cfreds_2015_data_leakage_pc.dd - 21 GB / 20 GiB

>[Proceed ] [ Quit ]

Note: Disk capacity must be correctly detected for a successful r
If a disk listed above has an incorrect size, check HD jumper set
detection, and install the latest OS patches and disk drivers.
```

```
/bin/bash 61x17
TestDisk 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk cfreds_2015_data_leakage_pc.dd - 21 GB / 20 GiB

Please select the partition table type, press Enter when done
>[Intel ] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax ] Humax partition table
[Mac ] Apple partition map (legacy)
[None ] Non partitioned media
[Sun ] Sun Solaris partition
[XBox ] Xbox partition
[Return ] Return to disk selection
```

```

/bin/bash 61x17
TestDisk 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk cfreds_2015_data_leakage_pc.dd - 21 GB / 20 GiB
  CHS 2611 255 63 - sector size=512Note: Correct disk geom
etry is required for a successful recovery. 'Analyse'process
[ Analyse ] Analyse current partition structure and search
> [ Advanced ] Filesystem Utils
[ Geometry ] Change disk geometry
[ Options   ] Modify options
[ MBR Code  ] Write TestDisk MBR code to first sector
[ Delete    ] Delete all data in the partition table
[ Quit      ] Return to disk selection

```

```

/bin/bash 61x17
TestDisk 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk cfreds_2015_data_leakage_pc.dd - 21 GB / 20 GiB - CHS 26
11 255 63
    Partition              Start      End      Size in s
  1 * HPFS - NTFS          0 32 33    12 223 19      2048
> 2 P HPFS - NTFS          12 223 20   2610 180 2      417341
44

[ Type ] [ Boot ] [ List ] > [Undelete] [Image Creat
ion]                               File undelete

```

- Use testDisk to recover files

```

TestDisk 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
2 P HPFS - NTFS      12 223 20 2610 180 2 41734144
Deleted files

MSMAPI\1033\mapisvc.inf          Previous          inode_77876          0
PyWinTypes27.dll                 inode_77876        110080
S-1-5-21-2425377081-3129163575-2985601102-1000/$I40295N          inode_77876          544
S-1-5-21-2425377081-3129163575-2985601102-1000/$I508CBB.jpg      inode_77876          544
S-1-5-21-2425377081-3129163575-2985601102-1000/$I55Z163          inode_77876          544
S-1-5-21-2425377081-3129163575-2985601102-1000/$I8YP3XK.jpg      inode_77876          544
S-1-5-21-2425377081-3129163575-2985601102-1000/$I9M7UMY          inode_77876          544
S-1-5-21-2425377081-3129163575-2985601102-1000/$ID0I3HE.jpg      inode_77876          544
S-1-5-21-2425377081-3129163575-2985601102-1000/$IFVCH5V.jpg      inode_77876          544
S-1-5-21-2425377081-3129163575-2985601102-1000/$II3FM2A.jpg      inode_77876          544
S-1-5-21-2425377081-3129163575-2985601102-1000/$IIQGWTT.ini      inode_77876          544
S-1-5-21-2425377081-3129163575-2985601102-1000/$IJEMT64.exe      inode_77876          544
S-1-5-21-2425377081-3129163575-2985601102-1000/$IKXD1U3.jpg      inode_77876          544
S-1-5-21-2425377081-3129163575-2985601102-1000/$IU3FKWI.jpg      inode_77876          544
S-1-5-21-2425377081-3129163575-2985601102-1000/$IX538VH.jpg      inode_77876          544
S-1-5-21-2425377081-3129163575-2985601102-1000/$IXWGVWC          inode_77876          544
S-1-5-21-2425377081-3129163575-2985601102-1000/$RIQGWTT.ini      inode_77876          174
>S-1-5-21-2425377081-3129163575-2985601102-1000/$RJEMT64.exe      inode_77876          0
Next
Use : to select the current file, a to select/deselect all files,
C to copy the selected files, c to copy the current file, q to quit

```

- Choose the destination and copy

```

TestDisk 7.1, Data Recovery Utility, July 2019

Please select a destination where the marked files will be copied.
Keys: Arrow keys to select another directory
      C when the destination is correct
      Q to quit

Directory /root/lab
>drwxr-xr-x      0      0      4096  4-Dec-2020 20:15 .
drwx-----      0      0      4096  4-Dec-2020 20:09 ..
drwxrwxrwx      0      0      4096 26-Nov-2020 16:03 AutomaticDestinations

```

Quit all

`ls S-1-5-21-2425377081-3129163575-2985601102-1000/ -l`

```

root@kali:~/lab#
root@kali:~/lab# ls S-1-5-21-2425377081-3129163575-2985601102-1000/ -l
total 64
-rw----- 1 root root 544 Mar 24 2015 '$I40295N'
-rw----- 1 root root 544 Mar 24 2015 '$I508CBB.jpg'
-rw----- 1 root root 544 Mar 24 2015 '$I55Z163'
-rw----- 1 root root 544 Mar 24 2015 '$I8YP3XK.jpg'
-rw----- 1 root root 544 Mar 24 2015 '$I9M7UMY'
-rw----- 1 root root 544 Mar 24 2015 '$ID0I3HE.jpg'
-rw----- 1 root root 544 Mar 24 2015 '$IFVCH5V.jpg'
-rw----- 1 root root 544 Mar 24 2015 '$II3FM2A.jpg'
-rw----- 1 root root 544 Mar 24 2015 '$IIQGWTT.ini'
-rw----- 1 root root 544 Mar 24 2015 '$IJEMT64.exe'
-rw----- 1 root root 544 Mar 24 2015 '$IKXD1U3.jpg'
-rw----- 1 root root 544 Mar 24 2015 '$IU3FKWI.jpg'
-rw----- 1 root root 544 Mar 24 2015 '$IX538VH.jpg'
-rw----- 1 root root 544 Mar 24 2015 '$IXWGVWC'
-rw----- 1 root root 174 Mar 24 2015 '$RIQGWTT.ini'
-rw----- 1 root root 26 Dec 13 1901 '$RJEMT64.exe:Zone.Identifier'
root@kali:~/lab#

```

- Match random file names with readable file names in the file

```

root@kali:~/lab# strings -h
Usage: strings [option(s)] [file(s)]
Display printable strings in [file(s)] (stdin by default)
The options are:
-a - --all             Scan the entire file, not just the data section [default]
-d - --data           Only scan the data sections in the file
-f - --print-file-name Print the name of the file before each string
-n - --bytes=[number] Locate & print any NUL-terminated sequence of at
<number>             least [number] characters (default 4).
-t - --radix={o,d,x}  Print the location of the string in base 8, 10 or 16
-w - --include-all-whitespace Include all whitespace as valid string characters
-o -                  An alias for --radix=o
-T - --target=<BFDNAME> Specify the binary file format
-e - --encoding={s,S,b,l,B,L} Select character size and endianness:
                        s = 7-bit, S = 8-bit, {b,l} = 16-bit {B,L} = 32-bit
-s - --output-separator=<string> String used to separate strings in output.
@<file>              Read options from <file>
-h - --help           Display this information
-v -V - --version     Print the program's version number
strings: supported targets: elf64-x86-64 elf32-i386 elf32-iamcu elf32-x86-64 pei-i386 pei-x86-6
4 elf64-l1om elf64-k1om elf64-little elf64-big elf32-little elf32-big pe-x86-64 pe-bigobj-x86-6
4 pe-i386 srec symbolsrec verilog tekhex binary ihex plugin
Report bugs to <http://www.sourceware.org/bugzilla/>
root@kali:~/lab# strings -el -f S-1-5-21-2425377081-3129163575-2985601102-1000/\$I508CBB.jpg
S-1-5-21-2425377081-3129163575-2985601102-1000/\$I508CBB.jpg: C:\Users\informant\AppData\Local\M
icrosoft\Windows\Burn\Burn\Hydrangeas.jpg
root@kali:~/lab#

```

Examine 'Recycle Bin' data in PC :

\$I Name	Timestamp Deleted	Original File (or Directory) Path
\$I40295N	2015-03-24 15:51:47	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\prop
???	???	???
???	???	???

2. What actions were performed for anti-forensics on PC at the last day '2015-03-25'?

Step 1.

- Search IE anti-forensics using keyword

```

root@kali:~/lab# cat webhistory/IE11.export/Container_1.4 | head -n 2
EntryId ContainerId CacheId UrlHash SecureDirectory FileSize Type F
lags AccessCount SyncTime CreationTime ExpiryTime Modified
Time AccessedTime PostCheckTime SyncCount ExemptionDelta Url F
ilename FileExtension RequestHeaders ResponseHeaders RedirectUrl Group E
xtendedData
24 1 0 46055402037931576 1 32988 69 1 1
Mar 23, 2015 18:12:08.184224700 Mar 23, 2015 18:12:08.084224600 Jan 01, 1601 00:
00:00.000000000 Feb 27, 2015 00:47:45.000000000 Mar 23, 2015 18:12:08.184224700
1 0 https://technet.microsoft.com/favicon.ico favicon[2].ico 4
85454502f312e3120323030204f4b0d0a436f6e74656e742d4c656e6774683a2033323938380d0a4
36f6e74656e742d547970653a20696d6167652f782d69636f6e0d0a455461673a202261346130343
230323735326430313a30220d0a5033503a2043503d22414c4c20494e442044535020434f5220414
44d20434f4e6f20435552204355536f204956416f204956446f20505341205053442054414920544
54c6f204f55522053414d6f20434e5420434f4d20494e54204e4156204f4e4c20504859205052452
050555220554e49220d0a582d506f77657265642d42793a204153502e4e45540d0a582d496e73746
16e63653a2043483131330d0a582d55412d436f6d70617469626c653a2049453d656467650d0a582
d506f77657265642d42793a204152522f322e350d0a5033503a2043503d22414c4c20494e4420445
35020434f522041444d20434f4e6f20435552204355536f204956416f204956446f2050534120505
344205441492054454c6f204f55522053414d6f20434e5420434f4d20494e54204e4156204f4e4c2
0504859205052452050555220554e49220d0a582d506f77657265642d42793a204153502e4e45540
d0a582d496e7374616e63653a2043483131330d0a0d0a 07000000000000010
root@kali:~/lab#

```

Three tasks:

1. Show all IE history
2. Grep date and with key word "anti"
3. Highlight anti in red

```

root@kali:~/lab# cat webhistory/IE11.export/Container* | grep -oEi "Mar 25, 2015.*search\?q=.*anti"
| grep anti --color
Mar 25, 2015 14:46:44.771005500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:46:44.753428500 Mar
25, 2015 14:46:44.752905500 Mar 25, 2015 14:46:44.771005500 0 0 0 Visited:
informant@http://www.bing.com/search?q=anti-forensic+tools&qsn&form=QBLH&pq=anti
Mar 25, 2015 14:46:44.752905500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:46:44.753428500 Mar
25, 2015 10:46:44.752000000 Mar 25, 2015 14:46:44.752905500 0 0 86400 :20150325
20150326: informant@http://www.bing.com/search?q=anti-forensic+tools&qsn&form=QBLH&pq=anti
Mar 25, 2015 14:46:44.752905500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:46:44.753428500 Mar
25, 2015 14:46:44.752905500 Mar 25, 2015 14:46:44.752905500 0 1 0 Visited:
informant@http://www.bing.com/search?q=anti-forensic+tools&qsn&form=QBLH&pq=anti
root@kali:~/lab#

```

- Search a list of anti-forensic keywords :

cat webhistory/IE11.export/Container* | grep -oEi "Mar 25, 2015, 2015.*search\?q=.*(steganography|encryp|cover|log|diab|trace|cover|remove|overwrite|de stroy|clean)"

```

root@kali:~/lab# cat webhistory/IE11.export/Container* | grep -oEi "Mar 25, 2015.
*search\?q=.*(steganography|encryp|cover|log|diab|trace|cover|remove|overwrite|d
estroy|clean)"
Mar 25, 2015 14:47:51.248205500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:47
:51.248728500 Mar 25, 2015 14:47:51.248205500 Mar 25, 2015 14:47:51.248205500
0 1 0 Visited: informant@http://www.bing.com/search?q=ccleaner&q
s=n&form=QBRE&pq=cclean
root@kali:~/lab#

```


Step 2.

- Search IE download history

Search key word “download”, only show first two records for the demonstration purpose :

```
cat webhistory/IE11.export/Container* | grep -oEi "Mar 25, 2015.*download" | head -n2 | grep download --color
```

```
root@kali:~/lab#  
root@kali:~/lab# cat webhistory/IE11.export/Container* | grep -oEi "Mar 25, 2015.*  
download" | head -n2 | grep download --color  
Mar 25, 2015 14:48:21.734705500 Mar 25, 2015 14:48:21.625505500 Mar 25, 2015 13:48:  
21.625505500 Jan 01, 1601 00:00:00.000000000 Mar 25, 2015 14:48:21.734705500 0 0  
0 http://www.piriform.com/ccleaner/download  
Mar 25, 2015 14:48:12.158005500 Mar 25, 2015 14:48:12.126805500 Mar 25, 2015 13:48:  
12.126805500 Jan 01, 1601 00:00:00.000000000 Mar 25, 2015 14:48:12.158005500 0 0  
0 http://www.piriform.com/ccleaner/download download  
root@kali:~/lab#
```

- Search key word “download”, only show last four records for the demonstration purpose

```
root@kali:~/lab#  
root@kali:~/lab# cat webhistory/IE11.export/Container* | grep -oEi "Mar 25, 2015.*  
download" | tail -n4 | grep download --color  
Mar 25, 2015 14:48:22.751205500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:41:  
13.254998900 Mar 25, 2015 14:48:22.751205500 Mar 25, 2015 14:48:22.751205500 0 1  
0 Visited: informant@http://www.piriform.com/ccleaner/download  
Mar 25, 2015 14:47:29.849805500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:47:  
29.850328500 Mar 25, 2015 14:47:29.849805500 Mar 25, 2015 14:47:29.849805500 0 1  
0 Visited: informant@http://sourceforge.net/projects/eraser/files/Eraser%206/  
6.2/Eraser%206.2.0.2962.exe/download  
Mar 25, 2015 14:47:05.844005500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:47:  
05.844528500 Mar 25, 2015 14:47:05.844005500 Mar 25, 2015 14:47:05.844005500 0 1  
0 Visited: informant@http://eraser.heidi.ie/download  
Mar 25, 2015 14:47:35.008505500 ie download M C:\\Users\\informant\\AppDa  
ta\\Roaming\\Microsoft\\Windows\\IEDownload  
root@kali:~/lab#
```

Step 3.

- Install download programs- shimcache

Check shimcache to monitor executed programs

```
root@kali:~/lab#  
root@kali:~/lab# rip.pl -r SYSTEM -p shimcache | grep 2015-03-25  
Launching shimcache v.20200428  
LastWrite Time: 2015-03-25 15:31:05Z  
C:\\Users\\informant\\Desktop\\Download\\Eraser 6.2.0.2962.exe 2015-03-25 14:47:40  
Executed  
C:\\Users\\INFORM~1\\AppData\\Local\\Temp\\eraserInstallBootstrapper\\dotNetFx40_Full  
_setup.exe 2015-03-25 14:50:15 Executed  
C:\\Users\\informant\\Desktop\\Download\\ccsetup504.exe 2015-03-25 14:48:28 Execu  
ted  
LastWrite Time: 2015-03-25 10:18:30Z  
root@kali:~/lab#
```

- Check if *eraser* or *ccleaner* is managed by Win installer

```

root@kali:~/lab# /bin/bash 8/x26
root@kali:~/lab# rip.pl -r SOFTWARE -p installer | grep -Ei "eraser|cclearn" -A 1 -B 2
Launching installer v.20200517
Key       : 1F782E6C74E20F54BB15498F51FCBF84
LastWrite: 2015-03-25 14:57:31Z
20150325 - Eraser 6.2.0.2962 6.2.2962 (The Eraser Project) ←
root@kali:~/lab#

```

- Check Software registry hive

`rip.pl -r SOFTWARE -p uninstall | grep 2015-03-25 -A 2 -B 2`

`rip.pl -r NTUSER_informant.DAT -p uninstall | grep 2015-03-25 -A 2 -B 2`

```

root@kali:~/lab# /bin/bash 86x26
root@kali:~/lab# rip.pl -r SOFTWARE -p uninstall | grep 2015-03-25 -A 2 -B 2
Launching uninstall v.20200525
Microsoft\Windows\CurrentVersion\Uninstall

2015-03-25 14:57:31Z
  Eraser 6.2.0.2962 v.6.2.2962 ←

2015-03-25 14:54:33Z
  Microsoft .NET Framework 4 Extended v.4.0.30319

2015-03-25 14:54:06Z
  Microsoft .NET Framework 4 Extended v.4.0.30319

2015-03-25 14:52:06Z
  Microsoft .NET Framework 4 Client Profile v.4.0.30319

2015-03-25 14:51:39Z
  Microsoft .NET Framework 4 Client Profile v.4.0.30319

2015-03-25 10:15:21Z
  DXM_Runtime
  MPlayer2
root@kali:~/lab# █

```

- Check *NTUSER_informant.DAT* registry hive

```

root@kali:~/lab# /bin/bash 86x26
root@kali:~/lab# rip.pl -r NTUSER_informant.DAT -p uninstall
Launching uninstall v.20200525
uninstall v.20200525
(Software, NTUSER.DAT) Gets contents of Uninstall keys from Software, NTUSER.DAT hives
Uninstall
root@kali:~/lab# █

```

Nothing found

- Check app paths registry

Check Software apppaths


```
root@kali:~/lab# rip.pl -r SOFTWARE -p apppaths | grep 2015-03-25 -A 2 -B 2
Launching apppaths v.20200511
(NTUSER.DAT,Software) Gets content of App Paths subkeys

2015-03-25 14:57:31Z
Eraser.exe - C:\Program Files\Eraser\Eraser.exe ←
2015-03-25 11:14:17Z
cmmgr32.exe -
2015-03-22 15:19:29Z
root@kali:~/lab#
```

Check NTUSER_informant apppaths

```
root@kali:~/lab# rip.pl -r NTUSER_informant.DAT -p apppaths
Launching apppaths v.20200511
apppaths v.20200511
(NTUSER.DAT,Software) Gets content of App Paths subkeys
```

Step 4

- Execute two programs : *userassist*

```
root@kali:~/lab# rip.pl -r NTUSER_informant.DAT -p userassist | grep 2015-03-25 -A 1 --color
Launching userassist v.20170204
2015-03-25 15:28:47Z
{1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\xpsrchvw.exe (1)
2015-03-25 15:24:48Z
{6D809377-6AF0-444B-8957-A3773F02200E}\Microsoft Office\Office15\WINWORD.EXE (4)
2015-03-25 15:21:30Z
{7C5A40EF-A0FB-4BFC-874A-C0F2E0B9FA8E}\Google\Drive\googledrivesync.exe (1)
2015-03-25 15:15:50Z
{6D809377-6AF0-444B-8957-A3773F02200E}\CCleaner\CCleaner64.exe (1)
2015-03-25 15:12:28Z
{6D809377-6AF0-444B-8957-A3773F02200E}\Eraser\Eraser.exe (1)
2015-03-25 14:57:56Z
C:\Users\informant\Desktop\Download\ccsetup504.exe (1)
2015-03-25 14:50:14Z
C:\Users\informant\Desktop\Download\Eraser 6.2.0.2962.exe (1)
2015-03-25 14:46:05Z
Microsoft.InternetExplorer.Default (5)
2015-03-25 14:42:47Z
Microsoft.Windows.MediaPlayer32 (1)
2015-03-25 14:41:03Z
{6D809377-6AF0-444B-8957-A3773F02200E}\Microsoft Office\Office15\OUTLOOK.EXE (5)
--
2015-03-25 15:21:30Z
{0139D44E-6AFE-49F2-8690-3DAFCAE6FFB8}\Google Drive\Google Drive.lnk (1)
2015-03-25 15:15:50Z
C:\Users\Public\Desktop\CCleaner.lnk (1) ←
2015-03-25 15:12:28Z
C:\Users\Public\Desktop\Eraser.lnk (1) ←
```

Step 5

Search for deleted .exe files

Only search for 2015-03-25 from Download folder

```
fls -rd -o 206848 cfreds_2015_data_leakage_pc.dd -l | grep -i Users/informant/Desktop/Download/ | grep 2015-03-25 --color
```

```
root@kali:~/lab#
root@kali:~/lab# fls -rd -o 206848 cfreds_2015_data_leakage_pc.dd -l | grep -i Users/informant/Desktop/Download/ | grep 2015-03-25 --color
-/r * 75101-128-4: Users/informant/Desktop/Download/Eraser 6.2.0.2962.exe 2015-03-25 10:47:40 (EDT) 2015-03-25 10:47:40 (EDT) 2015-03-25 10:47:40 (EDT) 8
0 (EDT) 2015-03-25 10:47:40 (EDT) 2015-03-25 10:47:40 (EDT) 2015-03-25 10:47:40 (EDT) 8
317032 0 0
-/r * 75101-128-5: Users/informant/Desktop/Download/Eraser 6.2.0.2962.exe:Zone.Identifier 20
15-03-25 10:47:40 (EDT) 2015-03-25 10:47:40 (EDT) 2015-03-25 10:47:40 (EDT) 2015-03-25
10:47:40 (EDT) 26 0 0
-/r * 75186-128-4: Users/informant/Desktop/Download/ccsetup504.exe 2015-03-25 10:48:28 (EDT) 2
015-03-25 10:48:28 (EDT) 2015-03-25 10:48:28 (EDT) 2015-03-25 10:48:28 (EDT) 53
44528 0 0
-/r * 75186-128-5: Users/informant/Desktop/Download/ccsetup504.exe:Zone.Identifier 2015-03-25
10:48:28 (EDT) 2015-03-25 10:48:28 (EDT) 2015-03-25 10:48:28 (EDT) 2015-03-25 10:48:2
8 (EDT) 26 0 0
root@kali:~/lab#
```

- Only search for deleted .exe from \$UsnJrnl on 2015-03-25

```
root@kali:~/lab#
root@kali:~/lab# grep -Ei "\.exe.*2015-03-25.*delete" UsnJrnl2Csv/UsnJrnl_2020-11-28_12-15-14.csv --color
0x03E0C358 Eraser 6.2.0.2962.exe 65061720|2015-03-25 14:47:40.1253055|CLOSE+FILE_DELETE|75101|2|22273|3|archive|2|0|0x00000000|0
0x03E0C8F8 Eraser%206.2.0.2962[1].exe 65063160|2015-03-25 14:47:40.2033055|CLOSE+FILE_DELETE|75100|1|70424|1|archive+not_indexed|2|0|0x00000000|0
0x03E19808 ccsetup504.exe 65116168|2015-03-25 14:48:28.8054055|CLOSE+FILE_DELETE|75186|1|22273|3|archive|2|0|0x00000000|0
0x03E19C80 ccsetup504[1].exe 65117312|2015-03-25 14:48:28.8678055|CLOSE+FILE_DELETE|75184|1|70423|1|archive+not_indexed|2|0|0x00000000|0
0x03E66AA8 TMPFF8D.tmp.exe 65432232|2015-03-25 14:50:47.3722055|CLOSE+FILE_DELETE|75477|3|75332|2|archive|2|0|0x00000000|0
0x03E66CD0 TMPFF8D.tmp.exe.tmp 65432784|2015-03-25 14:50:47.4658055|CLOSE+FILE_DELETE|75502|1|75332|2|directory+not_indexed|2|0|0x00000000|0
```

What actions were performed for anti-forensics on PC at the last day ‘2015-03-25’?

Timestamp	Behavior	Description
2015-03-25 10:46:44	Search anti-forensic methods	anti-forensic tools
???	???	???
???	???	???

YOU MUST SUBMIT A FULL-SCREEN IMAGE FOR FULL CREDIT!

Save the document with the filename "**YOUR NAME Lab 7.pdf**", replacing "YOUR NAME" with your real name.

Email the image to the instructor as an attachment to an e-mail message. Send it to: **xxx@fe.edu.vn** with a subject line of "**Lab 7 From YOUR NAME**", replacing "YOUR NAME" with your real name.